Chapter 1

Blockchain 101

Foundations of Blockchain Technology

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Based on: Mastering Blockchain, 2nd Edition (Packt Publishing)

Learning Objectives

- Define blockchain in both layman and technical terms
- Identify and explain the five key properties of blockchain
- Describe block structure and the generic elements of a blockchain
- Trace blockchain's historical origins and pre-Bitcoin foundations
- Distinguish between public, private, and permissioned blockchains
- Evaluate key benefits, limitations, and design trade-offs

Chapter Overview

Topics covered in Chapter 1 of Mastering Blockchain:

- Growth of Blockchain Technology
- Distributed Systems Foundations
- History of Blockchain & Electronic Cash
- Blockchain Defined — Layman & Technical
- Five Key Properties
- Block Structure & Generic Elements
- How Blockchain Accumulates Blocks
- Types and Tiers of Blockchain
- Consensus Mechanisms Overview
- CAP Theorem and Blockchain

Origins and Historical Context

Key Milestones

- 2008 — Satoshi Nakamoto publishes Bitcoin whitepaper
- 2009 — Genesis block mined; Bitcoin network launches
- 2013–2025 — Adoption: Ideas → Research → PoC → Production → Mainstream → Maturity
- 2018+ — Enterprise consortia: EEA, Hyperledger, R3
- Global blockchain spend projected at $9.2B by 2021

Pre-Bitcoin Foundations

- David Chaum — e-cash & blind signatures (1980s)
- Merkle trees — efficient transaction hashing
- Hash chains — cryptographic linkage
- Hashcash — computational puzzles (PoW precursor)
- P2P networks — decentralised communication
- Blockchain = convergence of existing technologies

Blockchain Defined

Layman's Definition

"Blockchain is an ever-growing, secure, shared record-keeping system in which each user holds a copy of the records, which can only be updated if all parties involved in a transaction agree to update."

Technical Definition

"Blockchain is a peer-to-peer, distributed ledger that is cryptographically-secure, append-only, immutable (extremely hard to change), and updateable only via consensus or agreement among peers."

Note: Blockchain ≠ Cryptocurrency. Bitcoin is one application of blockchain technology.

Five Key Properties of Blockchain

1. Peer-to-Peer	No central controller; all participants communicate directly — enables trustless transactions without intermediaries
2. Distributed Ledger	Ledger replicated across all network peers; each peer holds a complete and identical copy
3. Cryptographically Secure	Provides non-repudiation, data integrity, and data origin authentication via cryptography
4. Append-Only	Data added sequentially in time-ordered fashion; practically immutable; requires a 51% attack to alter past records
5. Updateable via Consensus	No central authority; updates validated by protocol agreement among peers — analogous to TCP/IP

How Blockchain Accumulates Blocks

1. Node creates a transaction and digitally signs it with its private key
2. Transaction broadcast via Gossip protocol; multiple nodes verify it independently
3. Verified transaction bundled into a block and broadcast — transaction receives its 1st confirmation
4. New block joins the ledger; the next block links back via a hash pointer — transaction gets 2nd confirmation
5. Each new block reconfirms all prior transactions — Bitcoin requires 6 confirmations for finality

Anatomy of a Block

Block Header

- Previous Block Hash — cryptographic link to prior block
- Timestamp — date and time of block creation
- Nonce — number adjusted during Proof of Work mining
- Merkle Root — root hash of all transactions in the block

Block Body

- List of validated transactions

Key Terms

- Genesis Block — first hardcoded block; no previous hash pointer
- Nonce — number used once; central to Proof of Work
- Merkle Root — enables efficient transaction verification
- Hash Pointer — link that makes any alteration immediately detectable
- Transaction — fundamental unit representing transfer of value

Types of Blockchain

Type	Access Model	Examples
Public	Open to all — permissionless	Bitcoin, Ethereum
Private	Invitation only — permissioned	HydraChain, Quorum
Consortium / DLT	Selected group — permissioned	Hyperledger, R3 Corda
Sidechain	Cross-chain asset transfer via Proof of Burn	Rootstock
Tokenless	No native cryptocurrency; data sharing only	Enterprise data platforms

Tiers of Blockchain Technology

Blockchain 1.0	Blockchain 2.0	Blockchain 3.0	Blockchain X.0
2009–2010	2010+	2012+	Future
Cryptocurrencies	**Smart Contracts**	**Beyond Finance**	**Singularity**
Digital currency & payments Bitcoin as the first application	Programmable blockchain Ethereum, Hyperledger	Government, health, arts, justice Decentralised applications (DApps)	Public service infrastructure AI agents (Machina Economicus)

Reference: Melanie Swan, Blockchain: Blueprint for a New Economy (O'Reilly, 2015)

Benefits and Limitations

BENEFITS

- Decentralisation — no trusted third party required
- Immutability — records extremely hard to alter
- High availability — P2P, no single point of failure
- Security — cryptographic data integrity
- Cost savings — no clearing house or intermediary fees
- Faster dealings — no reconciliation or lengthy verification

LIMITATIONS

- Scalability — low throughput vs. traditional systems
- Privacy — transparency conflicts with confidentiality needs
- Regulatory uncertainty — standards still evolving
- Immature technology — ecosystem still developing
- Adaptability — complex integration with legacy systems

Consensus Mechanisms

Algorithm	Mechanism	Used By	Key Trade-off
PoW	Computational puzzle	Bitcoin, Litecoin	High energy use
PoS	Coin stake as collateral	Ethereum, Peercoin	Energy-efficient
DPoS	Voting delegation	BitShares	Fast; less decentralised
PBFT	Voting rounds (BFT)	Hyperledger	Fast; limited scale
PoET	Intel TEE/SGX timer	Sawtooth Lake	Hardware dependency
FBA	Federated vote quorums	Stellar	Flexible trust model

Consensus is the backbone of blockchain — it provides decentralisation without a central authority

Key Terms and Definitions

Node
A participant in a distributed network; can be honest, faulty, or malicious

Byzantine Node
A network participant that behaves maliciously or irrationally

Genesis Block
The first hardcoded block in a blockchain; no previous block hash

Nonce
A number used once; central to Proof of Work mining and replay protection

Merkle Root
Root hash of a Merkle tree of all transactions in a block

Smart Contract
A self-executing program stored on the blockchain encoding business logic

51% Attack
When an attacker controls the majority of hash rate, enabling chain rewrite

CAP Theorem
A distributed system cannot simultaneously guarantee Consistency, Availability, and Partition Tolerance

Key Takeaways

- Blockchain is peer-to-peer, distributed, cryptographically-secure, append-only, and consensus-driven
- It originated with Bitcoin (2008) and is a convergence of existing technologies — not a single invention
- The five key properties work together as a system; none is revolutionary in isolation
- Three main categories: public (permissionless), private (permissioned), and consortium / DLT
- Consensus mechanisms balance decentralisation, speed, and energy use — no single algorithm suits all cases
- Blockchain trades strict consistency for availability + partition tolerance (CAP theorem)

Discussion & Q&A

1. Where do you see blockchain being most impactful in Indonesia over the next 5 years?

2. Is blockchain suitable for voting systems? What are the key trade-offs?

3. Why might an organisation choose a private blockchain over Bitcoin's public chain?

4. What tension exists between GDPR’s right to erasure and blockchain’s immutability?